

Matthew B. Luebbers

Postdoctoral Fellow
Georgia Institute of Technology
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Education

- University of Colorado Boulder**; Boulder, CO Aug 2018 – Aug 2024
Doctor of Philosophy, Computer Science
Advisor: Prof. Bradley Hayes
Collaborative AI and Robotics Lab (CAIRO)
Dissertation: *Spatially-Grounded Communication for Mental Model Alignment in Human-Robot Teams*
Committee: Profs. Bradley Hayes*, Alessandro Roncone, Daniel Szafir, Nisar Ahmed, & Michael Gleicher
- University of Colorado Boulder**; Boulder, CO Aug 2018 – May 2021
Master of Science, Computer Science
- Cornell University**; Ithaca, NY Aug 2014 – May 2018
Bachelor of Arts, Computer Science
Internal Concentration: Artificial Intelligence & Robotics
External Concentration: Psychology

Professional Experience

- Georgia Institute of Technology**; Atlanta, GA Sep 2024 – Present
Postdoctoral Fellow, School of Interactive Computing
Researching human-centric artificial intelligence and robotics in the Cognitive Optimization and Relational (CORE) Robotics Laboratory, advised by Prof. Matthew Gombolay.
- NASA Jet Propulsion Laboratory**; Pasadena, CA Jun 2023 – Aug 2023
Spacecraft Operations Engineering Intern, Instrument Data Systems (398)
Designed and developed VIBE (VR Image Browsing Extension), building off and improving the VR prototype I developed in the summer of 2019 for use driving the Mars rover Perseverance. Also participated in rover driving shifts for Perseverance and Curiosity, helping operate both vehicles on Mars.
- NASA Jet Propulsion Laboratory**; Pasadena, CA Jun 2020 – Aug 2020
Research Technologist Intern, Robotics (347)
Created the composite terrain generation pipeline for a Monte-Carlo simulation tool to enable statistical analysis of plans generated by the Perseverance rover's ENav system for autonomous navigation.
- NASA Jet Propulsion Laboratory**; Pasadena, CA Jun 2019 – Aug 2019
Robotic Systems Engineering Intern, Robotics (347)
Designed a prototype virtual reality application for stereo image viewing with 3D drive plan overlay for the Rover Sequencing & Visualization Program (RSVP). Intended for use in rover driving activities for the Mars rovers Curiosity and Perseverance. Also participated in tactical rover driving shifts for Curiosity, learning to sequence and operate the vehicle while helping command its drives to scientific targets.
- NASA Jet Propulsion Laboratory**; Pasadena, CA Jun 2018 – Aug 2018
Software Systems Engineering Intern, Instrument Data Systems (398)
Continued work on the Common Workflow Service (CWS), including creation of a web-based BPMN process modeler to contain all CWS functionality within a single webapp.
- NASA Jet Propulsion Laboratory**; Pasadena, CA Jun 2017 – Aug 2017
Software Systems Engineering Intern, Instrument Data Systems (398)
Continued work on the Common Workflow Service (CWS), including adaptation work for the Instrument Data Systems pipelines of the Mars 2020 mission.

Teaching Experience

CS 7648 (Interactive Robot Learning) Georgia Institute of Technology ; Atlanta, GA Guest Lecturer, Prof. Matthew Gombolay	Jan 2025 – May 2025
CS 3630 (Introduction to Robotics and Perception) Georgia Institute of Technology ; Atlanta, GA Guest Lecturer, Prof. Matthew Gombolay	Sep 2024 – Dec 2024
CSCI 5302 (Advanced Robotics) University of Colorado Boulder ; Boulder, CO Course Manager, Prof. Bradley Hayes	Aug 2021 – Dec 2021
CSCI 5722 (Computer Vision) University of Colorado Boulder ; Boulder, CO Teaching Assistant, Prof. Ioana Fleming	Jan 2020 – May 2020
CSCI 1300 (Introduction to Computer Science) University of Colorado Boulder ; Boulder, CO Teaching Assistant, Prof. Ioana Fleming	Aug 2019 – Dec 2019
CS 4700 (Foundations of Artificial Intelligence) Cornell University ; Ithaca, NY Teaching Assistant, Prof. Bart Selman	Aug 2017 – Dec 2017
CS 3410 (Computer System Organization & Programming) Cornell University ; Ithaca, NY Teaching Assistant, Prof. Anne Bracy	Aug 2016 – May 2017
CS 2110 (Object-Oriented Programming & Data Structures) Cornell University ; Ithaca, NY Course Consultant, Profs. David Gries, Nate Foster, & Ross Tate	Aug 2015 – May 2016

Publications

Asterisk (*) denotes shared first authorship

Journal Papers

- [J1] Aaquib Tabrez, Matthew B. Luebbers, and Bradley Hayes. (2020). **A Survey of Mental Modeling Techniques in Human-Robot Teaming**. In Current Robotics Reports. Springer-Nature.

Conference Papers

- [C1] Matthew B. Luebbers*, Aaquib Tabrez*, Kanaka Samagna Talanki, and Bradley Hayes. (2024). **Recency Bias in Task Performance History Affects Perceptions of Robot Competence and Trustworthiness**. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2024). Yokohama, Japan, *Acceptance Rate: 45%*.

- [C2] Yi-Shiuan Tung, Matthew B. Luebbers, Alessandro Roncone, and Bradley Hayes. (2024). **Workspace Optimization Techniques to Improve Prediction of Human Motion During Human-Robot Collaboration**. In Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2024). Boulder, Colorado, USA. **Best Paper Honorable Mention (Technical Track)**. *Acceptance Rate: 25%*.
- [C3] Matthew B. Luebbers*, Aaquib Tabrez*, Kyler Ruvane*, and Bradley Hayes. (2023). **Autonomous Justification for Enabling Explainable Decision Support in Human-Robot Teaming**. In Proceedings of Robotics: Science and Systems (RSS 2023). Daegu, South Korea. *Acceptance Rate: 31%*.
- [C4] Christine T. Chang, Matthew B. Luebbers, Mitchell Hebert, and Bradley Hayes. (2023). **Human Non-Compliance with Robot Spatial Ownership Communicated via Augmented Reality: Implications for Human-Robot Teaming Safety**. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2023). London, England, UK. *Acceptance Rate: 43%*.
- [C5] Matthew B. Luebbers*, Aaquib Tabrez*, and Bradley Hayes. (2022). **Descriptive and Prescriptive Visual Guidance to Improve Shared Situational Awareness in Human-Robot Teaming**. In Proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2022). Auckland, New Zealand. **Best Student Paper Runner-Up (Top 2 of 629 submissions)**. *Acceptance Rate: 26%*.
- [C6] Matthew B. Luebbers, Connor Brooks, Carl L. Mueller, Daniel Szafir, and Bradley Hayes. (2021). **ARC-LfD: Using Augmented Reality for Interactive Long-Term Robot Skill Maintenance via Constrained Learning from Demonstration**. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA 2021). Xi'an, China. *Acceptance Rate: 48%*.

Workshops, Symposia, & Posters

- [W1] Nina Moorman, Michelle Zhao, Matthew B. Luebbers, Sanne van Waveren, Reid Simmons, Henny Admoni, Sonia Chernova, and Matthew Gombolay. (2025). **Bi-Directional Mental Model Reconciliation for Human-Robot Interaction with Large Language Models**. In Proceedings of the Workshop on Advancing Artificial Intelligence through Theory of Mind (ToM4AI 2025). Philadelphia, Pennsylvania, USA.
- [W2] Matthew B. Luebbers and Bradley Hayes. (2024). **Explainable Guidance and Justification for Mental Model Alignment in Human-Robot Teams**. In Companion for the ACM/IEEE International Conference on Human-Robot Interaction (HRI Pioneers 2024). Boulder, Colorado, USA.
- [W3] Matthew B. Luebbers*, Aaquib Tabrez*, Kyler Ruvane*, Ashley H. Rabin, Kevin W. King, William Gerichs, and Bradley Hayes. (2024). **Hierarchical Multi-Agent Reinforcement Learning with Explainable Decision Support for Human-Robot Teams**. In Proceedings of the Workshop on Explainability for Human-Robot Collaboration (ExpHRC 2024). Boulder, Colorado, USA.
- [W4] Matthew B. Luebbers*, Yi-Shiuan Tung*, Alessandro Roncone, and Bradley Hayes. (2024). **Stereoscopic Virtual Reality Teleoperation for Human Robot Collaborative Dataset Collection**. In Proceedings of the Workshop on Virtual, Augmented and Mixed Reality for Human Robot Interaction (VAM-HRI 2024). Boulder, Colorado, USA.
- [W5] Matthew B. Luebbers*, Maciej K. Wozniak*, Max Pascher*, Bryce Ikeda*, and Ayesha Jena*. (2024). **Virtual, Augmented, and Mixed Reality for Human-Robot Interaction (VAM-HRI)**. In Companion for the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2024). Boulder, Colorado, USA.
- [W6] Carl L. Mueller, Matthew B. Luebbers, Aaquib Tabrez, and Bradley Hayes. (2023). **Augmented Reality and Proxy Grippers Improve Demonstration-based Robot Skill Learning**. In Proceedings of the Workshop on Life-Long Learning with Human Help (L3H2 2023). London, England, UK.
- [W7] Breanne Crockett*, Kyler Ruvane*, Matthew B. Luebbers, and Bradley Hayes. (2023). **Effective Human-in-the-loop Control Handover via Confidence-Aware Autonomy**. In Proceedings of the Workshop on Life-Long Learning with Human Help (L3H2 2023). London, England, UK.

- [W8] Yi-Shiuan Tung, Matthew B. Luebbers, Alessandro Roncone, and Bradley Hayes. (2023). **Improving Human Legibility in Collaborative Robot Tasks through Augmented Reality and Workspace Preparation**. In Proceedings of the Workshop on Virtual, Augmented and Mixed Reality for Human Robot Interaction (VAM-HRI 2023). Stockholm, Sweden.
- [W9] Matthew B. Luebbers*, Maciej K. Wozniak*, Christine T. Chang*, Bryce Ikeda*, Michael E. Walker*, Eric Rosen*, and Thomas Groechel*. (2023). **Virtual, Augmented, and Mixed Reality for Human-Robot Interaction (VAM-HRI)**. In Companion for the ACM/IEEE International Conference on Human-Robot Interaction (HRI 2023). Stockholm, Sweden.
- [W10] Matthew B. Luebbers*, Aaquib Tabrez*, and Bradley Hayes. (2022). **Augmented Reality-Based Explainable AI Strategies for Establishing Appropriate Reliance and Trust in Human-Robot Teaming**. In Proceedings of the Workshop on Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI 2022). Sapporo, Japan.
- [W11] Matthew B. Luebbers*, Christine T. Chang*, Aaquib Tabrez*, Jordan Dixon*, and Bradley Hayes. (2021). **Emerging Autonomy Solutions for Human and Robotic Deep Space Exploration**. In Proceedings of SpaceCHI: Human-Computer Interaction for Space Exploration (SpaceCHI 2021). Yokohama, Japan.
- [W12] Matthew B. Luebbers*, Aaquib Tabrez*, and Bradley Hayes. (2020). **Automated Failure-Mode Clustering and Labeling for Informed Car-To-Driver Handover in Autonomous Vehicles**. In Proceedings of the Workshop on Assessing, Explaining, and Conveying Robot Proficiency for Human-Robot Teaming. Cambridge, England, UK.
- [W13] Matthew B. Luebbers, Connor Brooks, Minjae John Kim, Daniel Szafir, and Bradley Hayes. (2019). **Augmented Reality Interface for Constrained Learning from Demonstration**. In Proceedings of the Workshop on Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI 2019). Daegu, South Korea.
- [W14] Matthew B. Luebbers, Ramchandran Muthukumar, Madeleine Udell, and Ross A. Knepper. (2017). **Planning Aerial Survey Missions using Low Rank Approximation**. Presented: Northeast Robotics Colloquium (NERC 2017). Boston, Massachusetts, USA.

Further Experience

Conference & Workshop Leadership

- Program Committee, Late Breaking Reports, “**ACM/IEEE International Conference on Human-Robot Interaction (HRI 2025)**.” Melbourne, Australia.
- Organizing Committee, “**Workshop on Athletic Robots and Dynamic Motor Skills (RoboLetics 2.0)**”, ICRA 2025, Atlanta, Georgia, USA.
- Organizing Committee, “**Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI)**”, HRI 2024. Boulder, Colorado, USA.
- Organizing Committee, “**Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI)**”, HRI 2023. Stockholm, Sweden.
- Organizing Committee, “**Exploring Applications for Autonomous Non-Verbal Human-Robot Interactions (nHRI)**”, HRI 2021. Cambridge, England, UK.

Honors and Awards

- **Best Paper Honorable Mention (Technical Track)**: HRI 2024 conference
- **Human-Robot Interaction (HRI) Pioneer**: 2024 cohort
- **Annual Research Expo Award (Best In-Progress Work)**: CU Boulder CS Dept., 2022/23 academic year
- **Best Student Paper Runner Up**: AAMAS 2022 conference (Top 2 of 629 submissions)
- **Annual Research Expo Award (Best Published Work)**: CU Boulder CS Dept., 2021/22 academic year
- **Dean’s Graduate Fellowship**: CU Boulder College of Engineering & Applied Science, 2018

Invited Talks & Panels

- **Models of Human Feedback for AI Alignment (MoFA)**. ICML Workshop Keynote Talk. Jul 2025.
- **Carnegie Mellon University**. Human-Robot Interaction Seminar. Mar 2025.
- **Cornell University**. Robotics Seminar (Special). Nov 2023.
- **Syracuse University**. Center for Advanced Systems and Engineering (CASE) Seminar. Nov 2023.
- **NASA Jet Propulsion Laboratory**. National Intern Day Panel. Jul 2023.
- **University of Colorado Boulder**. Robotics Summer Seminar Series. Aug 2022.

Conference & Journal Review

- Robotics and Automation Letters (RA-L)
- ACM Transactions on Human-Robot Interaction (THRI)
- IEEE/RSJ International Conference on Intelligent Robotics and Systems (IROS)
- IEEE International Conference on Robotics and Automation (ICRA)
- ACM/IEEE International Conference on Human-Robot Interaction (HRI)
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)

Research Mentoring

- **Marcus Lau**, *Masters student*, Georgia Institute of Technology
- **Yixing Yao**, *Masters student*, Georgia Institute of Technology
- **Nathan Howard**, *Masters student*, University of Colorado Boulder
- **Kyler Ruvane**, *Masters student*, University of Colorado Boulder
- **Kanaka Samagna Talanki**, *Masters student*, University of Colorado Boulder
- **Karthik Siddaramanna**, *Masters student*, University of Colorado Boulder
- **Aditi Periyannan**, *Undergraduate student*, Tufts University
- **Xinyu Cassie Cao**, *Masters student*, University of Colorado Boulder
- **Minjae John Kim**, *Undergraduate student*, University of Colorado Boulder

Additional Projects of Interest

Self-Driving RC Car

Aug 2019 – Dec 2019

Designed and built a self-driving RC car capable of racing around indoor tracks. Created in a team of five as a semester-long project for CSCI 5302 – Advanced Robotics.

Multiple Asteroid Flyby Mission Formulation

Jan 2019 – May 2019

Formulated a NASA Discovery-class mission concept for a robotic spacecraft to perform multispectral remote sensing of three main-belt asteroids to gain valuable understanding of planetary migration and solar system formation. Delivered a formal, cost-constrained proposal, and presented a PDR (Preliminary Design Review). Created in a team of ten as a semester-long project for ASEN 5148 – Spacecraft Design. I was responsible for the CDH (Command & Data Handling), and FSW (Flight Software) subsystems.

Autonomous Solar-Powered Airship

Sep 2016 – May 2018

Group project within Cornell's Robotic Personal Assistants Lab (RPAL) to design and build an airship capable of flying autonomously. A lightweight solar power system and robust autonomy extend the range of the airship, enabling it to perform long-duration aerial survey missions for ecological monitoring.